✅ 1. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- Highest relevance.
- This is fundamentally a geometric manipulation task:
  - Students must interpret a grid-based diagram as a scaled geometric figure.
  - They identify the boundary path of an irregular shape (the counter).
  - They count unit edges, handle corners, and avoid double-counting or missing segments.
- Additionally, they manipulate numerical data: converting grid units (e.g., 28 unit edges) to meters (28 × 0.5 = 14 m).
- This competency directly captures the core activity: working fluently with geometric and numerical representations.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving the problem requires a clear sequence of steps:
  1. Locate the counter on the plan.
  2. Trace its outer perimeter (not inner or other edges).
  3. Count the number of unit-length sides along this perimeter.
  4. Multiply by 0.5 m per unit to get total length.
  5. Present the final answer with units.
- Missing any step (e.g., confusing inner/outer edge, forgetting scale conversion) leads to error.
- This structured, multi-stage procedural reasoning is precisely what Competency #12 describes.

✅ 3. Competency #4: Make mathematical diagrams, graphs, constructions or computing artifacts

(and implicitly #6: Extract mathematical information from them)

- Highly relevant.
- Although students don't create a new diagram, they must actively use the given grid diagram as a mathematical tool.
- They extract geometric information (shape, boundary, segment counts) from the visual representation.
- Many will annotate the diagram (e.g., marking counted edges), effectively using it as a working artifact.
- Thus, the task engages both interpreting (#6) and utilizing (#4) mathematical diagrams—central to spatial problem-solving.

✅ 4. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Explicitly required.
- The prompt states: "Show your work."
- A strong response includes:
  - A clear count (e.g., "I counted 28 unit edges"),
  - The scale conversion (e.g., "Each unit = 0.5 m, so 28 × 0.5 = 14 m"),
  - Final answer with units ("Total edging needed: 14 metres").
- This demonstrates transparent communication of reasoning and intermediate steps, fulfilling Competency #8.

✅ 5. Competency #1: Perform a simple calculation

- Relevant but basic.
- The arithmetic itself is straightforward: multiplication and addition of small integers.
- However, it is embedded within a rich contextual and geometric setting.
- While necessary, this is the least sophisticated aspect—the real challenge lies in correctly identifying what to calculate.

✅ 6. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy

- Moderately relevant.
- The context informs key decisions:
  - Only the outer edge of the counter needs edging (not inner service side or walls).
  - Precision to the nearest 0.5 m is sufficient (since grid resolution is 0.5 m).
  - No need for decimal precision beyond one place.
- Students who ignore context might measure wrong edges or over-precise values.
- Thus, contextual awareness guides both interpretation and computational choices.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 这本质上是一道几何操作题：
  - 学生需将网格图理解为按比例缩放的几何图形；
  - 识别服务台的边界路径（一个不规则形状）；
  - 数出边界上的单位线段，处理转角，避免重复或遗漏。
- 同时还需处理数值数据：例如，若数出 28 条单位边，则换算为 28 × 0.5 = 14 米。
- 这项能力精准概括了本题核心：熟练操作几何与数值表示。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需遵循清晰的步骤序列：
  1. 在平面图中定位服务台；
  2. 沿其外边缘描出周长（非内侧或墙体）；
  3. 数出该周长包含多少个单位边；
  4. 乘以 0.5 米/单位，得到总长度；
  5. 给出带单位的最终答案。
- 若遗漏任一环节（如混淆内外边缘、忘记比例换算），就会出错。
- 这种结构化的多阶段程序性推理，正是能力 #12 所强调的。

✅ 3. 能力 #4：制作数学图表、图形、构造或计算产物

（以及隐含的能力 #6：从中提取数学信息）

- 高度相关。
- 虽然学生无需绘制新图，但必须主动将给定的网格图作为数学工具使用。
- 他们需从视觉表示中提取几何信息（形状、边界、线段数量）。
- 许多学生会在图上做标记（如勾选已数的边），实际上是在将图纸作为工作载体。
- 因此，该任务同时涉及解读（#6）和运用（#4）数学图表——这是空间问题解决的核心。

✅ 4. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 题目明确要求。
- 题干指出：“请展示你的计算过程。”
- 优秀回答应包括：
  - 明确的计数（如“我数出 28 条单位边”）；
  - 比例换算（如“每单位 = 0.5 米，所以 28 × 0.5 = 14 米”）；
  - 带单位的最终答案（“所需包边条总长：14 米”）。
- 这体现了透明地沟通推理过程与中间步骤，完全符合能力 #8。

✅ 5. 能力 #1：执行简单计算

- 相关但基础。
- 所涉算术很简单：小整数的加法与乘法。
- 然而，它嵌套在一个丰富的几何与情境背景中。
- 虽然必要，但这只是最基础的层面——真正的挑战在于正确判断“该计算什么”。

✅ 6. 能力 #13：利用对情境的理解来指导或加速数学解题过程，例如采用符合情境的精度

- 中等相关。
- 情境引导关键决策：
  - 只需在服务台的外边缘安装包边条（内侧或墙边不需要）；
  - 精度到 0.5 米 即可（因网格分辨率为 0.5 米）；
  - 无需保留多位小数。
- 忽略情境的学生可能测错边，或给出过度精确的答案。
- 因此，情境意识指导了从理解到计算的全过程。